



BAULKHAM HILLS HIGH SCHOOL

DECEMBER 2014
YEAR 12 TASK 1

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 50 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Section II
- Marks may be deducted for careless or badly arranged work

Total marks – 43

Exam consists of 5 pages.

This paper consists of TWO sections.

Section 1 – Page 2 (5 marks)

Questions 1-5

- Attempt Questions 1-5
- Allow about 5 minutes for this section

Section II – Pages 3-5 (38 marks)

- Attempt questions 6-8
- Allow about 45 minutes for this section

Topics Tested: Integration and Series with applications

Section I – Multiple choice questions (5 marks)

Use the multiple choice Answer Sheet for Question 1 – 5.

1. The first three terms of an arithmetic progression are 26, 23, 20. The sum S_n of the first n terms of the series is:

(A) $S_n = \frac{n}{2}(55 - 3n)$

(B) $S_n = 29 - 3n$

(C) $S_n = \frac{n}{2}(29 - 3n)$

(D) $S_n = 26 - 3n$

2. The definite integral of $\int_0^1 (5x^4 - x^2)dx =$

(A) $\frac{1}{15}$

(B) $\frac{2}{3}$

(C) 3

(D) $\frac{1}{2}$

3. The primitive function of $\sqrt{x} + 1$ is:

(A) $\frac{3}{2}\sqrt{x^3} + x + C$

(B) $\frac{3}{2}\sqrt{x^3} + x^2 + C$

(C) $\frac{2}{3}\sqrt{x^3} + x + C$

(D) $\frac{2}{3}\sqrt{x^5} + x + C$

4. The n^{th} term of the sequence 1, -2, 3, -4, 5, -6, 7, -8,..... is:

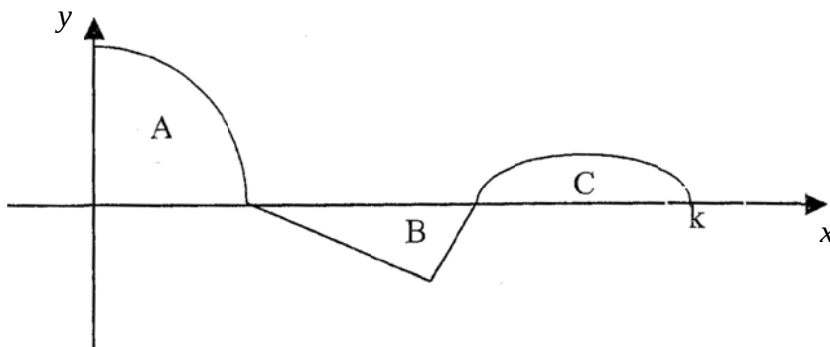
(A) $(-1)^{n-1}(2n - 1)$

(B) $(-1)^{n+1}n$

(C) $(-1)^{n+1}(2n - 1)$

(D) $(-1)^{n+1}(n + 1)$

5. The graph shows $y = f(x)$ for $0 \leq x \leq k$.



The value of $\int_0^k f(x) dx = 8.5$ units.

If area $A = 4$ units² and area $B = 3$ units², then area C is:

(A) 1.5

(B) 7.5

(C) 5.5

(D) 4.5

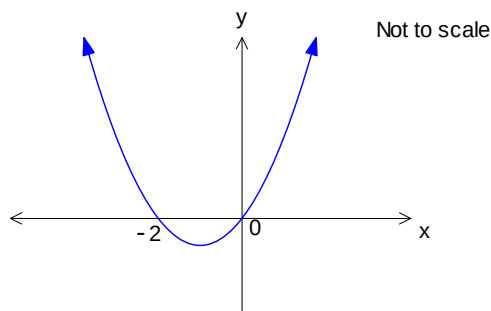
Section II – Extended response questions (38 marks)

Question 6 (12 marks) - Start a new page		
a)	Find the 40 th term of the arithmetic sequence 2, 6, 10...	2
b)	Find $\int (3x - 2)^5 dx$	2
c)	Evaluate $\sum_{k=1}^8 (4k - 1)$	2
d)	Evaluate $\int_2^3 \frac{x^2 + 5}{x^2} dx$	2
e)	Find the value(s) of x for which $(x - 1), (x + 3), (5x + 3)$ form a geometric sequence.	2
f)	Find $g(x)$, given $g'(x) = 3x^2 - 4$ and $g(1) = 4$.	2

Question 7 (13 marks) - Start a new page

- a) Find the area enclosed between the curve $y = x(x + 2)$ and the x axis.

2



- b) (i) Copy the table into your answer booklet and complete for the function $y = \frac{x-1}{x}$

1

x	1	2	3	4	5
y			$\frac{2}{3}$		

- (ii) Hence use Simpson's Rule with five function values to find the approximate value of $\int_1^5 \frac{x-1}{x} dx$ to three decimal places.

2

- c) The sum of the second and the fifth term of an arithmetic sequence is 32 whilst the sum of the third and the eighth term is 48.

- (i) Find the first term and the common difference.

2

- (ii) Find the sum of the first 30 terms.

2

- d) The sum of the first n terms of a sequence is given by $S_n = 75n - 2n^2$.

Find

- (i) The 2th term.

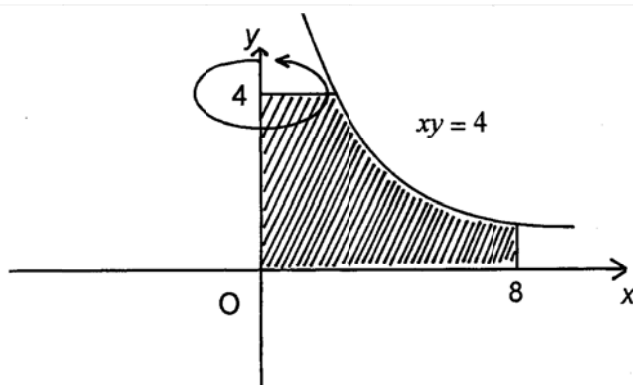
2

- (ii) The n^{th} term.

2

Question 8 (13 marks) - Start a new page

a)



The area enclosed by the curve $xy = 4$, the x and y axes and the lines $y = 4$ and $x = 8$, is rotated about the y axis.

3

By considering this area as two regions, find the volume of the solid.

b) Jessica has decided that she needs to set up a superannuation fund for her retirement. Her financial advisor told Jessica that she needs \$700,000 in the fund when she retires in 25 years' time.

To achieve this, she decides to make equal payments of \$ P at the beginning of each year. The fund pays an interest rate of 8% pa compounded annually.

Let A_n be the account balance at the end of n years, before she makes her payment for the following year.

(i) Show that her account balance after 3 years (before making the 4th payment) is given by:

2

$$A_3 = P(1.08)((1.08)^2 + (1.08) + 1)$$

(ii) Show that her account balance when she retires after 25 years is given by:

2

$$A_{25} = 13.5P \times (1.08^{25} - 1).$$

(iii) Hence find the amount Jessica will need to pay each year to satisfy her retirement requirements.

2

c) Consider the geometric series

$$1 + (\sqrt{\alpha} - 2) + (\sqrt{\alpha} - 2)^2 + (\sqrt{\alpha} - 2)^3 + \dots$$

(i) Find the largest positive integral value of α for the series to have a limiting sum.

2

(ii) Find the limiting sum for this series when $\alpha = 8$.

2

Express your answer as a surd with a rational denominator.

END OF EXAM

Section I - Multiple choice questions (5 marks)
Use the multiple choice Answer Sheet for Question 1 - 5.

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(A) $S_n = \frac{n}{2}(55 - 3n)$ (B) $S_n = 29 - 3n$
(C) $S_n = \frac{n}{2}(29 - 3n)$ (D) $S_n = 26 - 3n$

2. The definite integral of $\int_0^1 (5x^4 - x^2) dx =$

(A) $\frac{1}{15}$ (B) $\frac{2}{3}$
(C) 3 (D) $\frac{1}{2}$

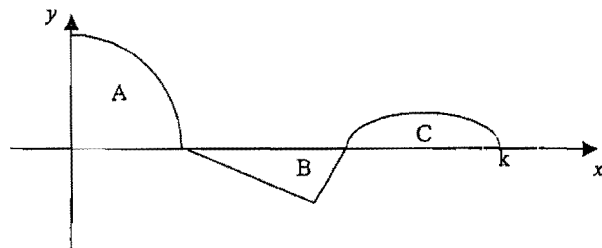
3. The primitive function of $\sqrt{x} + 1$ is:

(A) $\frac{3}{2}\sqrt{x^3} + x + C$ (B) $\frac{3}{2}\sqrt{x^3} + x^2 + C$
(C) $\frac{2}{3}\sqrt{x^3} + x + C$ (D) $\frac{2}{3}\sqrt{x^3} + x + C$

4. The n^{th} term of the sequence 1, -2, 3, -4, 5, -6, 7, -8, ... is:

(A) $(-1)^{n-1}(2n-1)$ (B) $(-1)^{n+1}n$
(C) $(-1)^{n+1}(2n-1)$ (D) $(-1)^{n+1}(n+1)$

5. The graph shows $y = f(x)$ for $0 \leq x \leq k$.



The value of $\int_0^k f(x) dx = 8.5$ units.

If area A = 4 units² and area B = 3 units², then area C is:

(A) 1.5 (B) 7.5
(C) 5.5 (D) 4.5

Y12 Advanced Maths

Q6. a) $a=2$ $d=6-2=4$ ①
both a and d.

$$T_{40} = a + (n-1)d$$

$$= 2 + 39 \times 4$$

$$= 158$$

b) $\int (3x-2)^5 dx$

$$= \frac{(3x-2)^{5+1}}{5+1} \times \frac{1}{3} + C$$

$$= \frac{(3x-2)^6}{18} + C$$

c) $\sum_{k=1}^8 (4k-1) = (4 \times 1 - 1) + (4 \times 2 - 1) + \dots + (4 \times 8 - 1)$

$$S = 136$$

$a=3, d=4, n=8$

d) $\int_2^3 \frac{x^2+5}{x^2} dx = \int_2^3 1 + \frac{5}{x^2} dx$

$$= \left[x - \frac{5}{x} \right]_2^3$$

$$= \left(3 - \frac{5}{3} \right) - \left(2 - \frac{5}{2} \right)$$

$$= \frac{11}{6}$$

$$e) \frac{x+3}{x-1} = \frac{5x+3}{x+3}$$

$$x^2+6x+9 = 5x^2+3x-5x-3$$

$$4x^2-8x-12=0$$

$$\therefore x^2-2x-3=0 \quad (1)$$

$$(x-3)(x+1)=0$$

$$x=3 \text{ or } x=-1$$

(1)

$$8) \int (3x^2-4) dx = x^3-4x+C \quad (1)$$

$$g(1) = 1^3-4 \times 1 + C = 4$$

$$\therefore C=7$$

$$g(x) = x^3-4x+7 \quad (1)$$

$$(Q7) a) y = x(x+2) = x^2+2x$$

$$A = \left| \int_{-2}^0 (x^2+2x) dx \right|$$

$$= \left| \left(\frac{x^3}{3} + x^2 \right) \right|_{-2}^0 \quad (1)$$

$$= \left| 0 - \left(\frac{(-2)^3}{3} + (-2)^2 \right) \right|$$

$$= \left| +\frac{8}{3} - 4 \right| = \frac{4}{3} \text{ u}^2. \quad (1)$$

b) i)

x	1	2	3	4	5
y	0	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{3}{4}$	$\frac{4}{5}$

(1)

ii)

$$\int_1^5 \frac{x-1}{x} dx = \int_1^3 \frac{x-1}{x} dx + \int_3^5 \frac{x-1}{x} dx$$

$$= \frac{3-1}{6} \left[f(1) + 4f\left(\frac{1+3}{2}\right) + f(3) \right] \quad (1)$$

$$+ \frac{5-3}{6} \left[f(3) + 4f\left(\frac{3+5}{2}\right) + f(5) \right]$$

$$= \frac{1}{3} \left[0 + 4 \times \frac{1}{2} + \frac{2}{3} \right] + \frac{1}{3} \left[\frac{2}{3} + 4 \times \frac{3}{4} \right]$$

$$= \frac{1}{3} \left(2 + \frac{2}{3} \right) + \frac{1}{3} \left(\frac{2}{3} + 3 + \frac{4}{3} \right)$$

$$= \frac{1}{3} \times \left[\frac{8}{3} + \frac{10 + 45 + 12}{15} \right]$$

$$= \frac{1}{3} \left[\frac{8}{3} + \frac{67}{15} \right]$$

$$= \frac{1}{3} \times \frac{107}{15}$$

$$= \frac{107}{45} \quad (1)$$

$$= 2.378 \quad \leftarrow (3 \text{ dec.})$$

c) i) $T_2 + T_5 = 32$

$$T_3 + T_8 = 48$$

$$(a+d) + (a+4d) = 2a+5d = 32$$

$$(a+2d) + (a+7d) = 2a+9d = 48$$

Eqs (2) - (1) $4d = 16$
 $\therefore d = 4 \quad (1)$

$$2a + 5d = 32$$

$$2a + 5 \times 4 = 32$$

$$\therefore a = 6 \quad (1)$$

ii) $S_{30} = \frac{30}{2} (2 \times 6 + (30-1) \times 4) \quad (1)$
 $= 1920 \quad (1)$

d) $S_n = 75n - 2n^2$

i) $S_1 = 75 \times 1 - 2 \times 1^2 = 73.$

$$S_2 = 75 \times 2 - 2 \times 2^2 = 150 - 8 = 142. \quad (1)$$

$$T_2 = S_2 - S_1 = 142 - 73 = 69 \quad (1)$$

ii) $S_{n-1} = 75(n-1) - 2(n-1)^2$
 $= 79n - 2n^2 - 77 \quad (1)$

$$T_n = S_n - S_{n-1}$$

$$= 75n - 2n^2 - (79n - 2n^2 - 77)$$

$$= -4n + 77 \quad (1)$$

Q8)

a)

$x = 8, y = \frac{1}{2}$

$xy = 4$
 $x = \frac{4}{y}$
 $x^2 = \frac{16}{y^2}$
 $= 16y^{-2}$

$V = \pi r^2 h + \pi \int_{\frac{1}{2}}^4 x^2 dy$
 $= \pi \times 8^2 \times \frac{1}{2} + \pi \int_{\frac{1}{2}}^4 16y^{-2} dy$
 $= 32\pi + \pi \int_{\frac{1}{2}}^4 \frac{16}{y^2} dy$
 $= 32\pi + \pi \left[-16y^{-1} \right]_{\frac{1}{2}}^4$
 $= 32\pi + \pi \left[-4 - (-32) \right]$
 $= 60\pi \text{ units} \quad 28\pi$
or 188.50 units (2 decs)

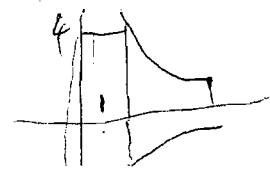
If rotating about x-axis (Incorrect).

$y = 4, xy = 4$
 $x = \frac{4}{y} = 1$

$xy = 4$
 $y = \frac{4}{x}$
 $y^2 = \frac{16}{x^2}$

2 marks

If all correct



$V = \pi r^2 h + \pi \int_1^8 y^2 dx$

$= \pi \times 4^2 \times 4 + \pi \int_1^8 \frac{16}{x^2} dx$
 $= 64\pi + \pi \left[-\frac{16}{x} \right]_1^8$
 $= 64\pi + \pi \left[-\frac{16}{8} + \frac{16}{1} \right]$
 $= 64\pi + 14\pi = 78\pi$

(30 marks)

b) $r = 8\%$

i) Beginning of Y1.

Balance $B = P$

Interest $I = P \times 8\%$

End of Y1
Balance $B = P + P \times 8\% = 1.08P$ ①

Beginning of Y2
Balance $B = P + 1.08P$

End of Y2
Interest $I = (P + 1.08P) \times 8\%$
Balance $B = (P + 1.08P) + (P + 1.08P) \times 8\%$

$= (P + 1.08P)(1 + 0.08)$

$= (P + 1.08P)(1.08)$

$= P \times 1.08 + P \times (1.08)^2$

$A_2 = P \times 1.08(1.08 + 1)$ ①

Similarly end of Y3

$A_3 = P \times 1.08(1.08^2 + 1.08 + 1)$

ii) Similarly after 25th Year

$A_{25} = P \times 1.08(1.08^{24} + 1.08^{23} + \dots + 1.08 + 1)$
 $= P \times 1.08 \left(\frac{1.08^{25} - 1}{1.08 - 1} \right)$ ① GP

$= P \times 1.08 \times \frac{(1.08^{25} - 1)}{0.08}$

$a = 1$
 $r = 1.08$
 $n = 25$

$$P \times 1.08 \times \frac{(1.08^{25} - 1)}{0.08} = 700000 \quad (1)$$

$$P = \frac{700000 \times 0.08}{1.08(1.08^{25} - 1)}$$

$$= \$8865.88 \text{ per year} \quad (1)$$

$$c) \quad 1 + (\sqrt{x}-2) + (\sqrt{x}-2)^2 + (\sqrt{x}-2)^3 + \dots$$

$$r = \frac{T_2}{T_1} = \frac{\sqrt{x}-2}{1} = \sqrt{x}-2 \quad \alpha \geq 0$$

$$\frac{T_3}{T_2} = \frac{(\sqrt{x}-2)^2}{(\sqrt{x}-2)} = \sqrt{x}-2$$

It has limiting sum when $|\sqrt{x}-2| < 1 \quad (1)$

$$\text{or } -1 < \sqrt{x}-2 < 1$$

$$-1+2 < \sqrt{x} < 1+2$$

$$1 < \sqrt{x} < 3$$

$$\text{or } 1^2 < x < 3^2$$

$$1 < x < 9$$

$\therefore$ The largest positive integer of x is $x=8 \quad (1)$

ii)

$$S_{\infty} = \frac{a}{1-r} = \frac{1}{1-(\sqrt{8}-2)} \quad (1)$$

$$= \frac{1}{1-\sqrt{8}+2}$$

$$= \frac{1}{3-\sqrt{8}}$$

$$= \frac{1 \cdot (3+\sqrt{8})}{(3-\sqrt{8})(3+\sqrt{8})}$$

$$= \frac{3+\sqrt{8}}{3^2-8} = \frac{3+\sqrt{8}}{9-8} = 3+\sqrt{8} \quad (1)$$

$$= 3+2\sqrt{2}$$

$$\frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} - \frac{1}{10} + \frac{1}{11} - \frac{1}{12} + \frac{1}{13} - \frac{1}{14} + \frac{1}{15} - \frac{1}{16} + \frac{1}{17} - \frac{1}{18} + \frac{1}{19} - \frac{1}{20} + \frac{1}{21} - \frac{1}{22} + \frac{1}{23} - \frac{1}{24} + \frac{1}{25} - \frac{1}{26} + \frac{1}{27} - \frac{1}{28} + \frac{1}{29} - \frac{1}{30} + \frac{1}{31} - \frac{1}{32} + \frac{1}{33} - \frac{1}{34} + \frac{1}{35} - \frac{1}{36} + \frac{1}{37} - \frac{1}{38} + \frac{1}{39} - \frac{1}{40} + \frac{1}{41} - \frac{1}{42} + \frac{1}{43} - \frac{1}{44} + \frac{1}{45} - \frac{1}{46} + \frac{1}{47} - \frac{1}{48} + \frac{1}{49} - \frac{1}{50} + \frac{1}{51} - \frac{1}{52} + \frac{1}{53} - \frac{1}{54} + \frac{1}{55} - \frac{1}{56} + \frac{1}{57} - \frac{1}{58} + \frac{1}{59} - \frac{1}{60} + \frac{1}{61} - \frac{1}{62} + \frac{1}{63} - \frac{1}{64} + \frac{1}{65} - \frac{1}{66} + \frac{1}{67} - \frac{1}{68} + \frac{1}{69} - \frac{1}{70} + \frac{1}{71} - \frac{1}{72} + \frac{1}{73} - \frac{1}{74} + \frac{1}{75} - \frac{1}{76} + \frac{1}{77} - \frac{1}{78} + \frac{1}{79} - \frac{1}{80} + \frac{1}{81} - \frac{1}{82} + \frac{1}{83} - \frac{1}{84} + \frac{1}{85} - \frac{1}{86} + \frac{1}{87} - \frac{1}{88} + \frac{1}{89} - \frac{1}{90} + \frac{1}{91} - \frac{1}{92} + \frac{1}{93} - \frac{1}{94} + \frac{1}{95} - \frac{1}{96} + \frac{1}{97} - \frac{1}{98} + \frac{1}{99} - \frac{1}{100}$$